University of Information Technology & Sciences (UITS)

Faculty of Science and Engineering

Department of Computer Science and Engineering

Program: B.Sc. in CSE

Term Final Examination, Autumn 2024

Course Title: Digital Logic Design

Course Code: CSE 215

Marks: 50 **Time: 3*(three)* hours**

(Answer all questions)

1. a) Implement a Full Adder by using two 4 x1 MUX. [03]

   b) Explain canonical form and convert the following to other canonical forms. [03]
      i. $F(A,B,C,D) = \sum(0,2,5,6,8,9,11,14)$
      ii. $F(w,p,q) = \Pi(0,3,7)$

   c) Convert the expression into sum of product and product of sum. [04]
      $F(A,B,C,D) = (AB + C)(B + C'D)$

2. a) Simplify the following Boolean function by using k – map [05]
      i. $F(A,B,C,D) = \Sigma(0,1,2,4,5,6,8,9,12,13,14)$
      ii. $F(w,x,y,x) = \Pi(1,3,5,7,13,15)$

   b) Simplify the Boolean function F using the don't- care conditions d, in i) sum of minterm and ii) product of maxterm using K map. [05]
      $F(A,B,C,D) = B'DE' + A'BE + B'C'E' + A'BC'D'$
      $d(A,B,C,D) = BDE' + CD'E'$
      $BCD' + A'BD + B'C'D'$ / $BCD' + BC'D'$

3. a) Implement a full adder with two half adders and an OR gate. Also prove that the complement of XOR is equal to XNOR. [04]

   b) Construct a 5-to-32-line decoder with four 3-to-8-line decoders with enable and one 2-to-4-line decoder and explain how it will work. [03]

   c) An Encoder will produce a binary code equivalent to the minterm, which is active High with enable. Design an 8 to 3 encoder with a truth table and circuit diagram. [03]

4. a) The following three Boolean functions specify a combinational circuit: [05]
      $$F_1(x,y,z) = (x' + y)z$$
      $$F_2(x,y,z) = (y'z' + x'y + yz')$$
      $$F_3(x,y,z) = (x + y)z$$
      Implement the circuit with a decoder constructed with OR gates. Use a block diagram for the decoder. Minimize the number of inputs in the external gates.

   b) State Flip Flop and construct T Flip Flop by using JK F/F. [05]

5. a) Design a sequential circuit from the following state diagram. (use JK F/F) [10]

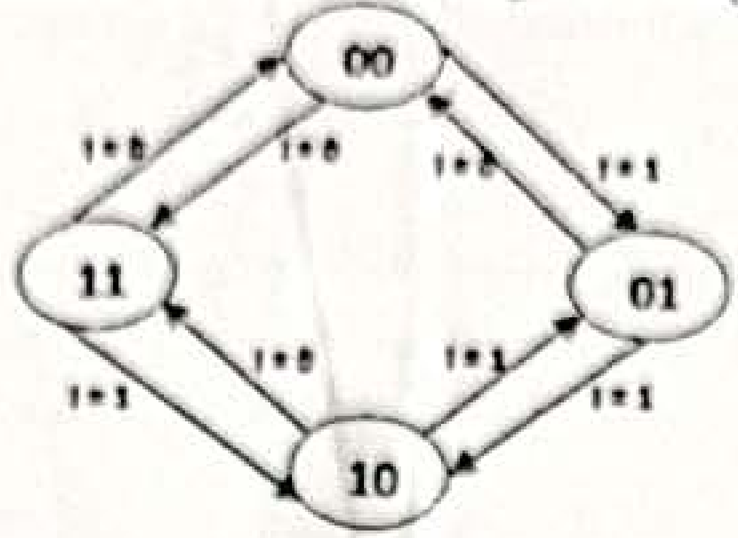

Figure 5(a)